COMP 3900 Computer Science Project 2023 T1

Project report

Project title: Movie Finder System

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1.Overview

1.1 Application Architecture

To achieve our team's goals, we have decided to implement a three-tiered system consisting of the Presentation layer, Business layer, and Data layer. Here's a brief description of each layer:

The Presentation layer is responsible for creating an intuitive user interface (UI) that facilitates interaction between users and the application. It presents data in a meaningful way and processes user input before passing it on to the Business layer.

The Business layer contains all the business logic necessary for running the application smoothly. This includes rules and algorithms that define how everything works together. The Presentation Layer sends requests to this tier which then processes data before sending it over to the Data Layer.

Data Layer is the layer that manages storage and retrieval of all data used by our application. It interacts with databases or other sources as needed by processing requests from both Presentation and Business layers.

By combining these three layers into one cohesive unit, we can create a modular architecture that separates concerns within our app. This allows developers to modify or replace individual layers without affecting others, making maintenance easier while ensuring security/performance overall, and becoming easier to debug. This layering approach has largely resulted in higher development efficiency.

1.2 System Structure

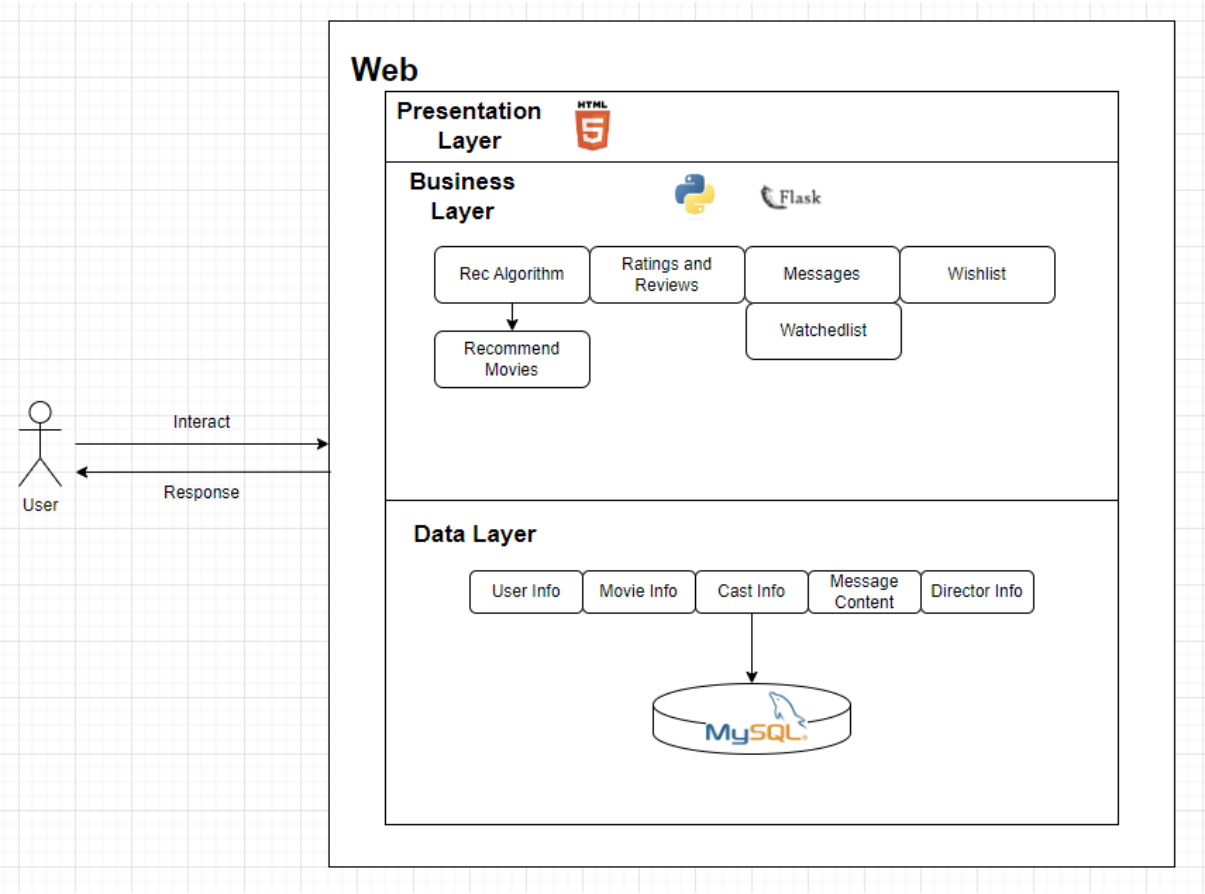


Figure 1.2.1

1.2.1 Presentation Layer

We use ReactJS framework as our presentation layer. ReactJS is a popular open-source JavaScript library for building user interfaces. It was developed by Facebook and is now maintained by a community of developers.

ReactJS follows a declarative approach to programming, which means that you tell it what you want the interface to look like and ReactJS will handle the underlying logic of how it should be rendered. ReactJS uses a virtual DOM (Document Object Model) to efficiently update and render changes to the interface. We use it to contribute the frontend.

1.2.2 Business Layer

Our business layer is developed based on python and Flask framework, which is a lightweight Python web framework built on Werkzeug and Jinja2 libraries. Through this framework, we can read the data stored in the database, and display the data through various interfaces.

Using this framework allows user input from the frontend to be transmitted to the backend, and then returns the results generated by the backend to the frontend.

1.2.3 Data Layer

For data storage, we use the MySQL database, which is very stable, has high availability and reliability, and supports large-scale data processing. Especially for a large amount of data like movie information, it is very friendly for dealing with the data. MySQL can run on a variety of operating systems, such as Windows, Linux, and Mac OS X. This makes our system highly compatible.

```
mysql> show tables;
ERROR 4031 (HY000): The client was disconnected by the server beca
use of inactivity. See wait_timeout and interactive_timeout for co
nfiguring this behavior.
No connection. Trying to reconnect...
Connection id:    1596
Current database: moviefinder

+------------------------+
| Tables_in_moviefinder |
+------------------------+
| chat_tbl               |
| direct_in_tbl          |
| director_tbl           |
| movie_review_tbl       |
| movie_tbl              |
| movie_watchlist_tbl    |
| movie_wishlist_tbl     |
| star_in_tbl            |
| star_tbl               |
| token_tbl              |
| user_blacklist_tbl     |
| user_friendship_tbl    |
| user_tbl               |
+------------------------+
13 rows in set (0.14 sec)
```

Figure 1.2.2.1

As you can see in the Figure 1.2.2.1, our specific information about movies is stored in a table named ‘movie_tbl’. For other functions of this system, we scattered it into other tables. In this way, various data in the database can be easily accessed.

1.3 Page Structure

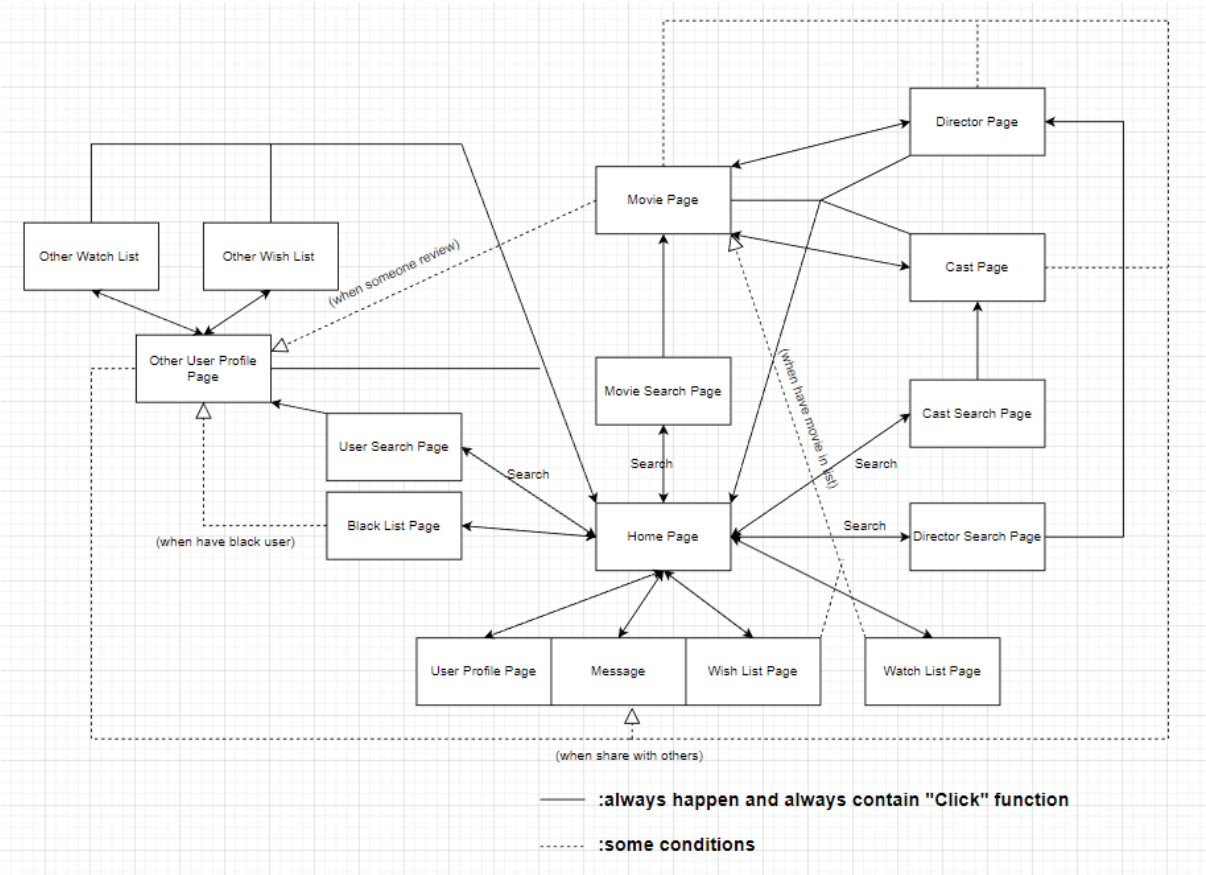


Figure 1.3.1

Each box represents a specific webpage, and each line represents a click or selection. Figure 1.3.1 provides a summary of the structure of the entire webpage, which helps to form a macroscopic view of the whole website.

1.4 Project Objectives

1.4.1 User stories:

Role	User Story	Acceptance Criteria
User	I can sign in, log in, log out, change the password, and forget the password so that I can manage my own account. [user login system]	The system allows users to create a new account for themselves using their E-mail. Users can log in with E-mail and password. User can change their passwords after they logged in. The system can help user to reset the password through email authentication when they forget the password.
User	I can browse personal information and modify	User information includes email, first name, last name, age, sex,

	personal information at any time so that I can let other users know about me. [user communication system]	occupation, area, description, tag, permission level. For new accounts that have just been created, some user information is empty. Users can update their personal information excludes tag and permission level in their account detail page.
User	I can use the system without signing in, but I cannot leave a review. So that more people can use this system. [user login system]	The system allows user to search the movie, see the review, see the detail of the movie, see the detail of the director, see the detail of the cast and see other users' account detail.
User	I need to have a Wishlist so that I can add and remove any movies from the Wishlist. [blacklist, Wishlist, watched list system]	User must be able to add and remove any movie to the Wishlist. User can be able to view other users' Wishlist in their information page.
User	I need a blacklist so that I cannot see comments from users on the Blacklist. [blacklist, Wishlist, watched list system]	Users must be able to add and remove users from the Blacklist.
User	I can mark the movie as already seen. The details page of the movie should show how many users want to see and have already seen it. This can help me quickly determine whether the movie is popular or not. [blacklist, Wishlist, watched list system]	User must be able to add and remove any movie they come across in the system to a watched list. The details page of the movie should show how many users want to see and have already seen it.
User	I can view the wish list and already seen list of any other movie finder that has left a review for a given movie so that I can find some movies that I am potentially	User can be able to view other users' Wishlist and already seen list in their information page.

	interested in. [user communication system]	
User	I can share the movie, director, and user with another user, so that I can have better communication with others. [user communication system]	Each pair of friends can chat with each other and share movies, directors, casts, and users.
Movie finder	I need to see all the details about the specific movie that I am looking for, including its name, description, genre, cast, director, latest average rating, screenshots, and all associated reviews. [movie basic information search system]	Movie information includes movie name, release date, director, language, description, tag, list of casts, visit volume, rating, review, screen shot. List of casts is sorted by their participation.
Movie finder	I can search for the movie that I am interested in by keywords. It can show a list of movies when the keywords match their name, description, director, or genre, so I can find the specific movie that I am looking for. The list can be sorted. [movie basic information search system]	Once the keyword exactly matches part of the movie name, tag, director name and cast name, the movie should add into the list. The result list can be sorted by release date, alphabetical order, rating, number of reviews, number of views.
Movie finder	I need to have a specific page about a cast. [cast basic information search system]	Information of cast includes name, birthday, sex, nation, views, number of performances, pictures.
Movie finder	I can see a page about a director which shows all the information about her/him. So, I can know better about her/him. [director basic information search system]	Information of director includes name, description, birthday, nation, sex, visit volume, number of performances.
Movie finder	I can leave or delete reviews	User must also be able to leave a

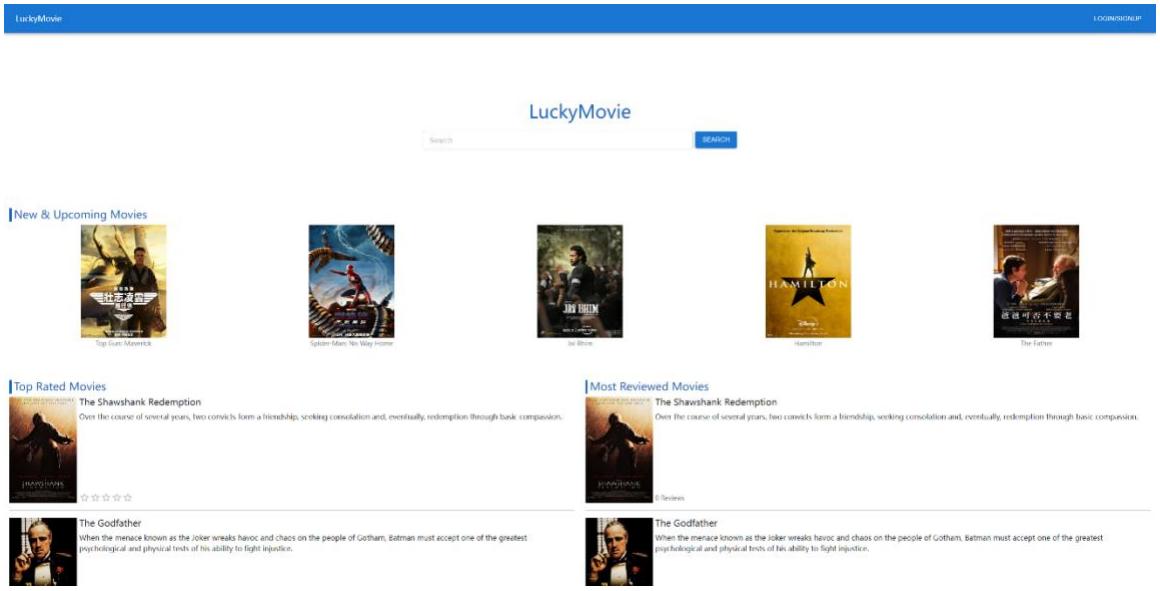
	for any given movie so that other users can see these reviews and ratings. The review includes text and rating. [movie rating and review system]	review for any given movie they view, which includes text and a rating from 0 to 5. The review displayed on the detail page of the movie should include content, rating, release date, number of agree, number of disagree, author.
Movie finder	I can have a search without specific this search for a user, movie, or director. So that I can have a better experience using this system. [movie basic information search system]	User can choose whether to search for a movie, cast, director or user after the search results are available. On the movie search results page, each movie is displayed with the movie name, description, rating and number of reviews. On the cast search results page there are cast names, descriptions, views, and number of performances. On the director search results page there is the name of the director, description, views, number of performances. On the user search results page there is the name of the user, views, and the number of friends. The movie result list can be sorted by release date, alphabetical order, rating, number of reviews, number of visits. The cast result list can be sorted by alphabetical order, views, number of performances. The director result list can be sorted by alphabetical order, views, numbers of performances. The user result list can be sorted by alphabetical order, views, numbers of friends.
Movie finder	I want to see some other movies that are "similar" on the full details page, so that I	In the page of movie detail, there should be a part for the “similar” movie. The movies in this section

	can see some movies that I may be potentially interested in. [movie recommendation system]	should be relevant to the user's wish list and the movie they are currently viewing. There also should be a you_may_like movie, which recommend some movies you may like.
Product owner	I can delete malicious comments from users so that I can manage this software better. [management system for product owner]	Administrators have the right to delete comments in order to prevent malicious comments from leading public opinion. For the normal development of the system and the film industry, administrators have higher rights than ordinary users.
Product owner	I can Add labels to specific users, modify the information of users who do not conform to the specifications, and cancel the number of prohibited users so that I can manage this software better. [management system for product owner]	Some users may post comments pretending to be influencers, so it is necessary that only the administrator can add the user's authoritative tag.
Developer	I should make the search function support forced inclusion and exclusion so that users can have more accurate search results. [movie basic information search system]	If the user's input in the search box contains a "-‘ ’", the default search result must not contain the word between the "’" and “”. If the user's input in the search box contains a "‘ ’", the default search result must contain the word between the "’" and “”. Search system is case-insensitive.
Developer	I need to improve the safety of the movie finder system and protect the users' privacy. [website security system]	The website should be able to withstand a sudden influx of requests for access.

2.Functionalities

2.1 Home

On the Home page, we provide real-time recommendations for New & Upcoming Movies, Top Rated Movies, and Most Reviewed Movies.

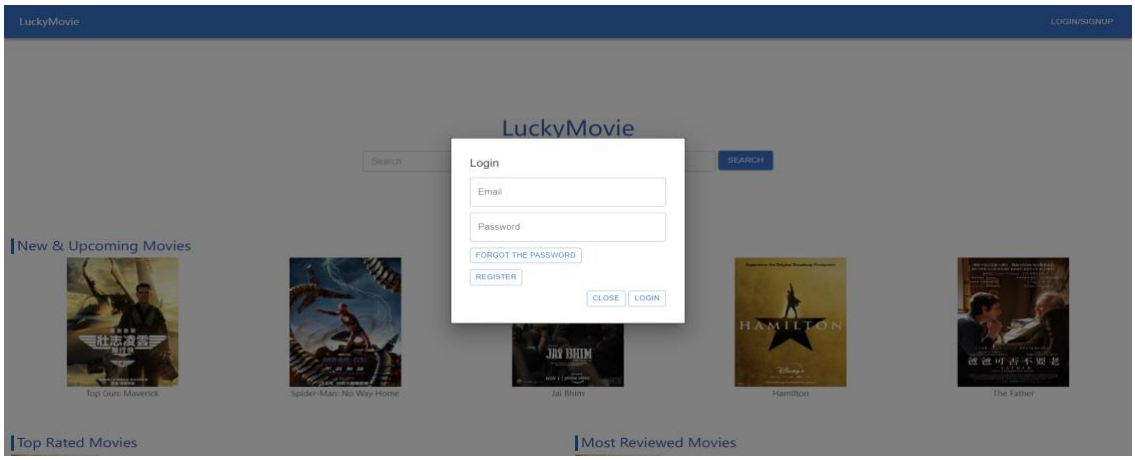


2.2 User Account

2.2.1 Login

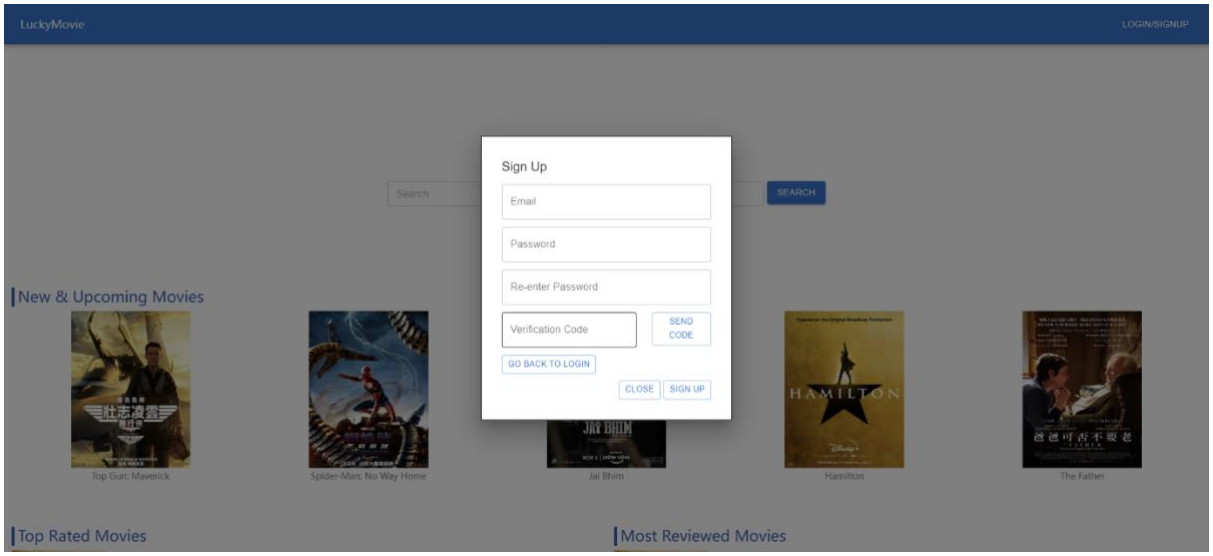
There is a login/register button in the upper right corner of the top bar, click it to log in or register. If you are a visitor, this button will always remain the same. When you log in, the status bar will elegantly showcase a plethora of options including profile page, Wishlist, watched list, blacklist, message and logout.

Conditions	Results
Email or Password is incorrect or invalid.	A relevant error message will appear
Account has not been email-validated	A relevant error message will appear
Click “login/signup”	Show the signin popup
Click “Login”	Successfully login
Click “close”	Close this popup
Click “register”	Navigate to the registration page.
Click forgot the password	Navigate to the reset password page.



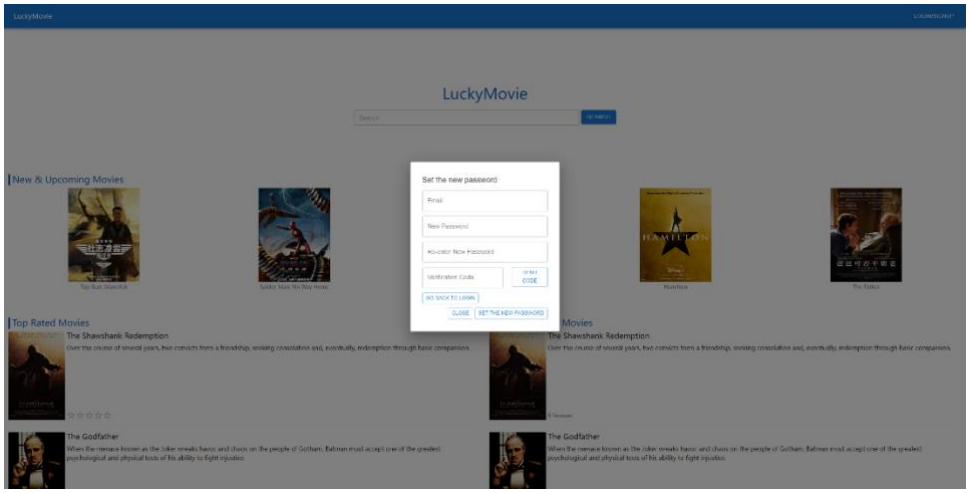
2.2.2 Register

Conditions	Results
Email format should be correct.	Display a relevant error message
The length of password should be longer than 6.	Display a relevant error message
The passwords entered twice do not match.	Display a relevant error message
Click the “Send code”	A verification email will be sent to the user containing a link that will activate the account
Click “SIGH UP”	Successfully registered
Click “go back to login”	Return to the signin popup
Click “close”	Close this popup



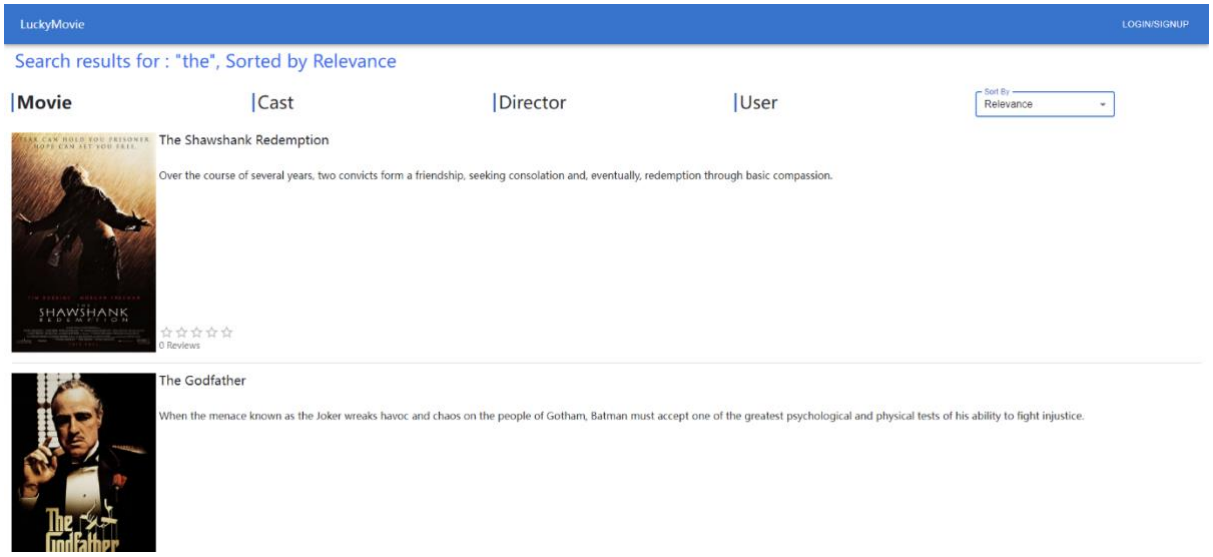
2.2.3 Set New Password

Condition	Result
Email format should be correct.	Display a relevant error message
The length of password should be longer than 6.	Display a relevant error message
The passwords entered twice do not match.	Display a relevant error message



2.3 Search Page

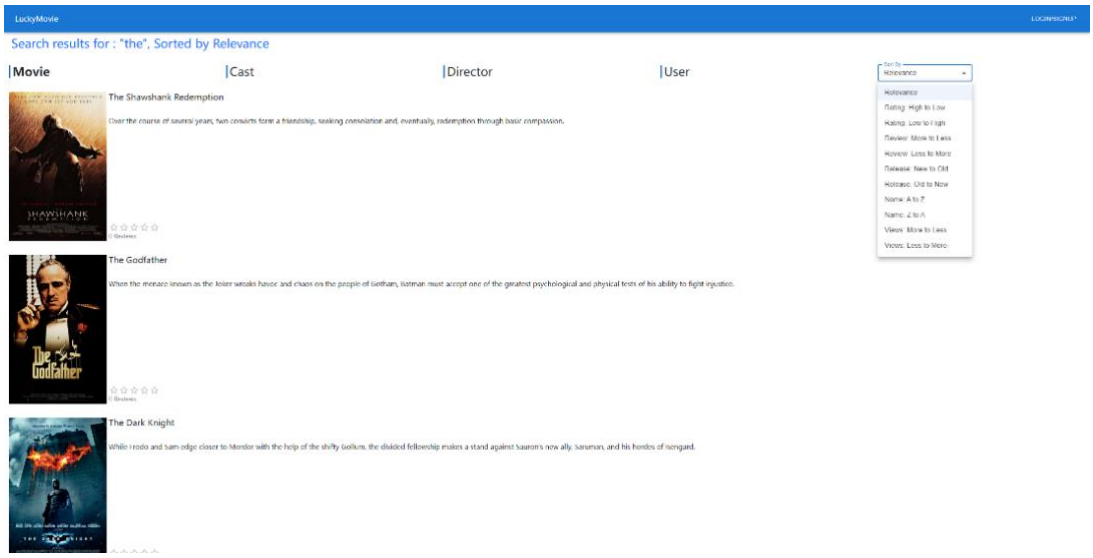
Condition	Result
Click “Movie”	If the movie information contains keywords, the movie will be displayed
Click “User”	If the user information contains keywords, the user will be displayed
Click “Cast”	If the cast information contains keywords, the user will be displayed
Click “Director”	If the director information contains keywords, the user will be displayed
Click “more”	Will show the whole description
Click “sort by”	You can choose different criteria to sort



2.3.1 Search Movie

Users can search for movies by the movie's name, category, director, and actors. They can also sort by relevance, rating, review number, release date, name and views number. When clicking on any movie title, they will be directed to the details page of that movie.

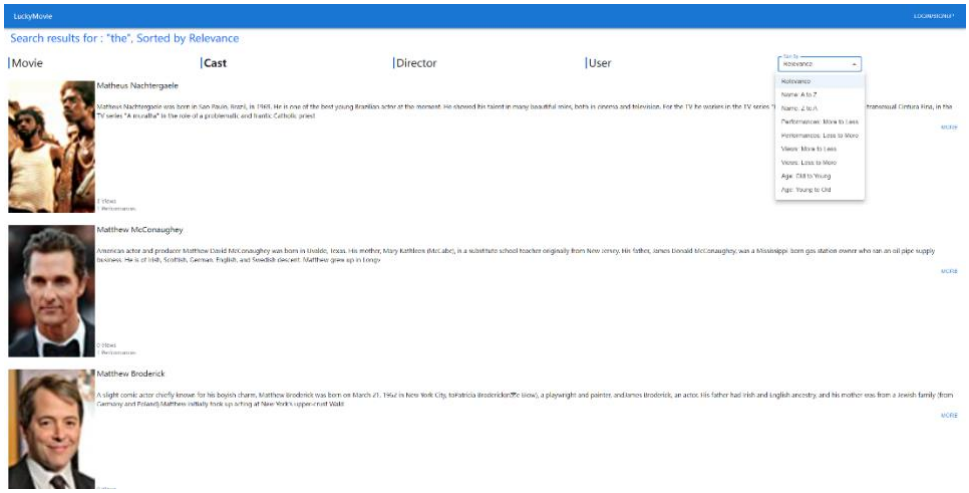
Conditions	Results
Click “more”	Will show the whole description
Click the movie	it will jump to the movie page



2.3.2 Search Cast

Users can search for actors by their name or the movies they have appeared in. They can also sort the results by relevance, name, number of performances, views and age. If an actor's description is too long, there will be a 'more' button that displays the full description when clicked. Clicking on any actor will take you to their details page.

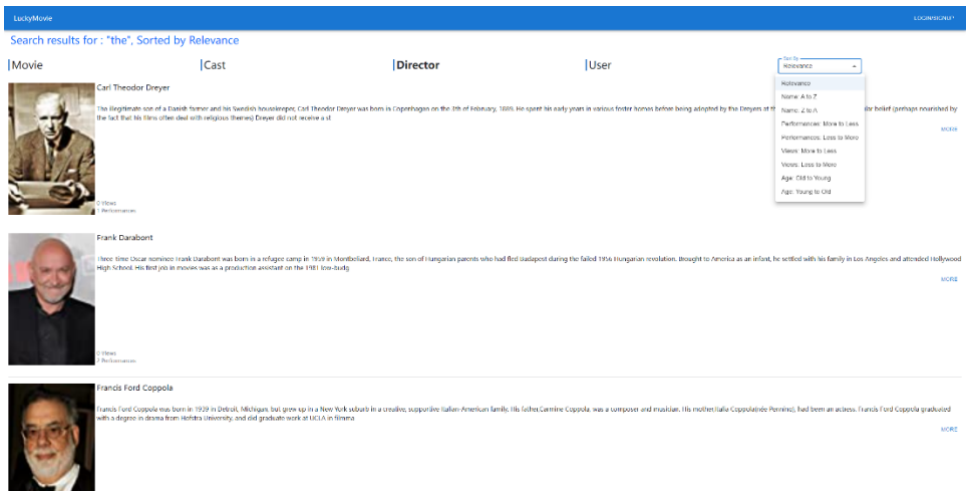
Conditions	Results
Click “more”	Will show the whole description
Click the cast	it will jump to the cast profile page



2.3.3 Search Director

Users can search for directors by their name, the movies they filmed. You can also sort by Relevance, Name, Performances number, Views number and Age. When the director's description is too long, there will be a “more” button, and the complete description will be displayed after clicking. When clicking on any director, it will jump to the director's details page.

Conditions	**Results**
Click “more”	Will show the whole description
Click the director	it will jump to the director profile page

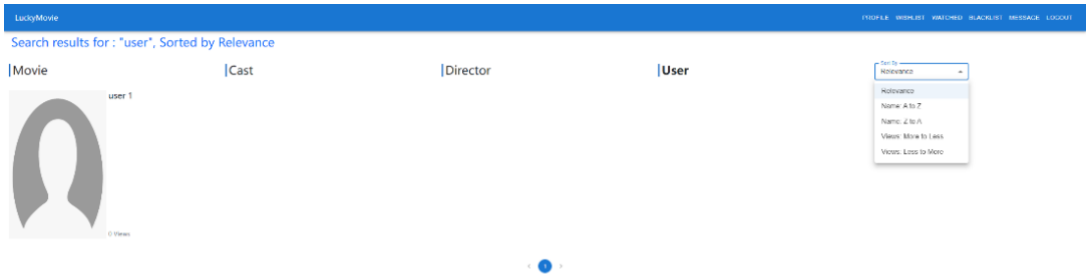


2.3.4 Search User

Users can be identified by the user's first name. It can also be ordered according to Relevance, Name and Views number.

Conditions	**Results**

Click “more”	Will show the whole description
Click the user	it will jump to the user's profile page

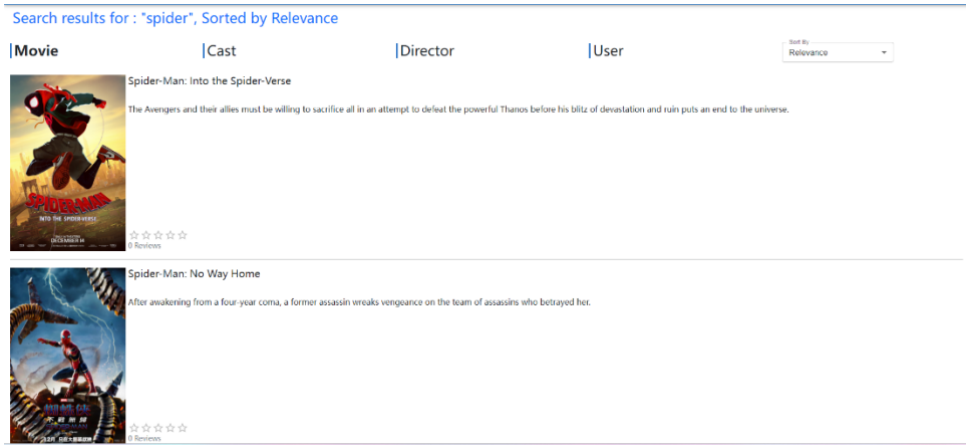


2.3.5 Advanced Search Movie

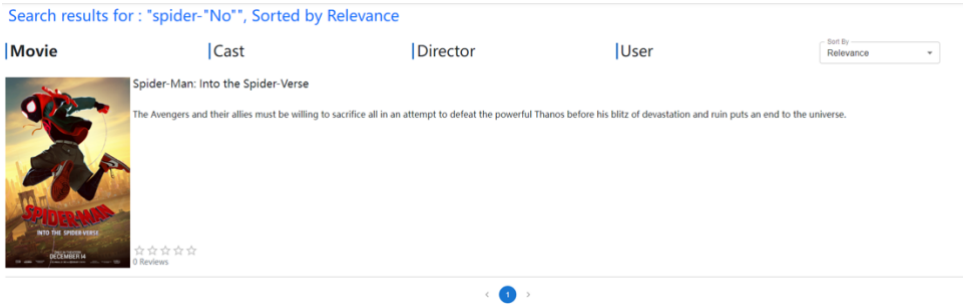
Description: Based on the movie search function mentioned in 2.3.1 above, we added the exclusion of terms when searching. When a user does a movie search and wants to exclude certain words from the search results, the movie containing that term will no longer be shown to the user in result page.

Operation method: search the words you are interested in - " Words you don't want to show up in search results".

For example: Search for movies based on 2.3.1， When users want to search for movies about spiders, it will display:



Then for Advanced Search Movie ： When user wants to search for movies about spiders, but they don't want to show up movies with "No" in the movie name on the search results page. The user types of spiders- " No" and the page will display:

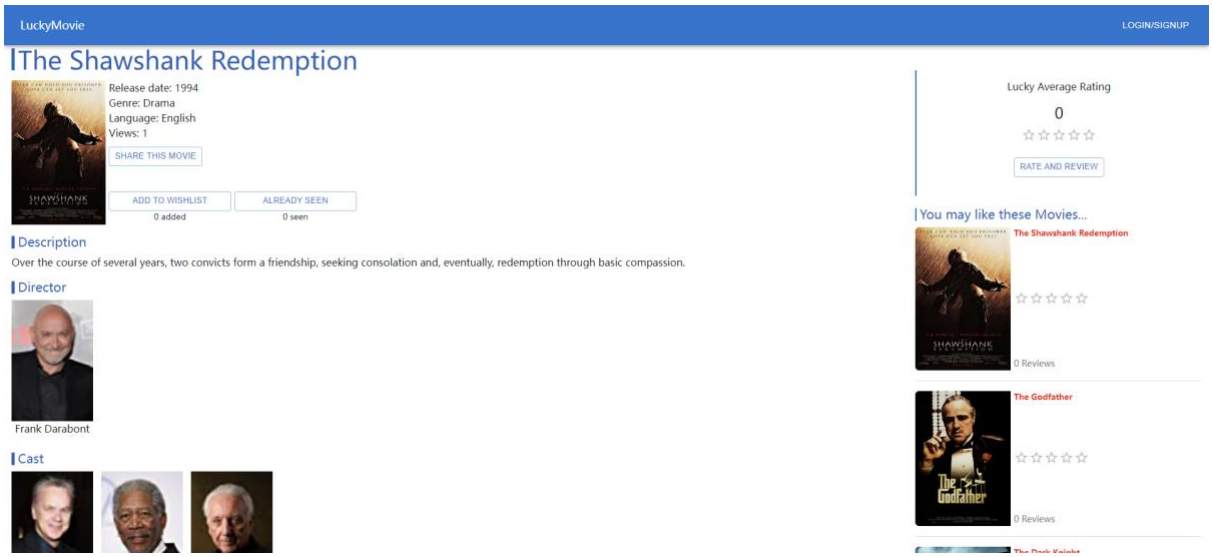


Note: Only if "" contains the exact same word in movie title will not be displayed in the movie search results.

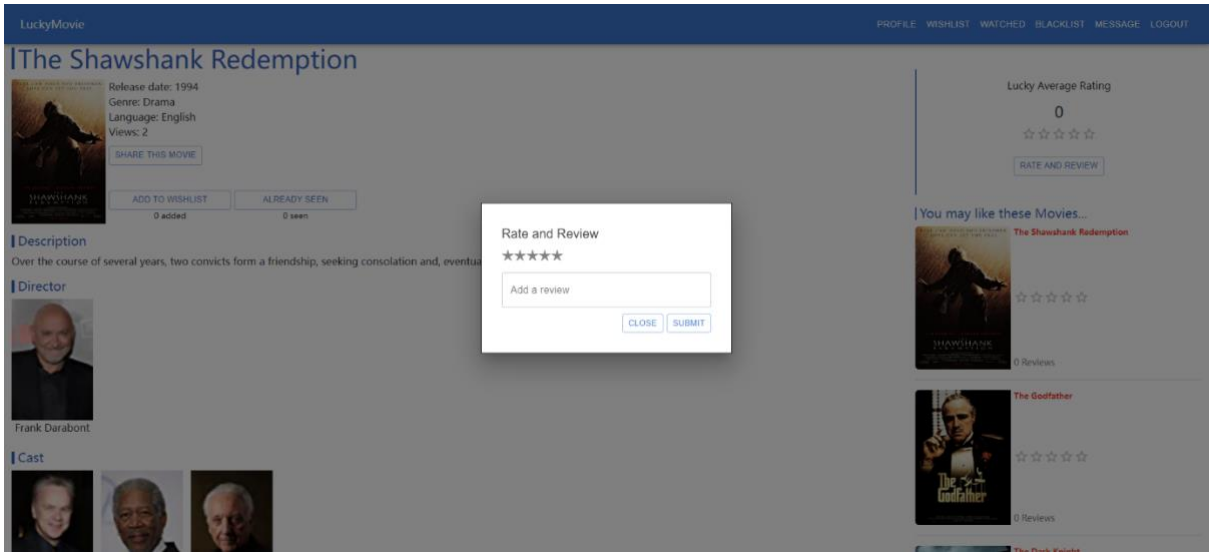
2.4 Movies Page

The movie details page contains information such as the actor list, our website site rating and the movie's description. Users can add this movie to their wishlist, watched list.
Users can also leave a review containing their ratings and comment. Recommendations for the current movie are displayed at the right.

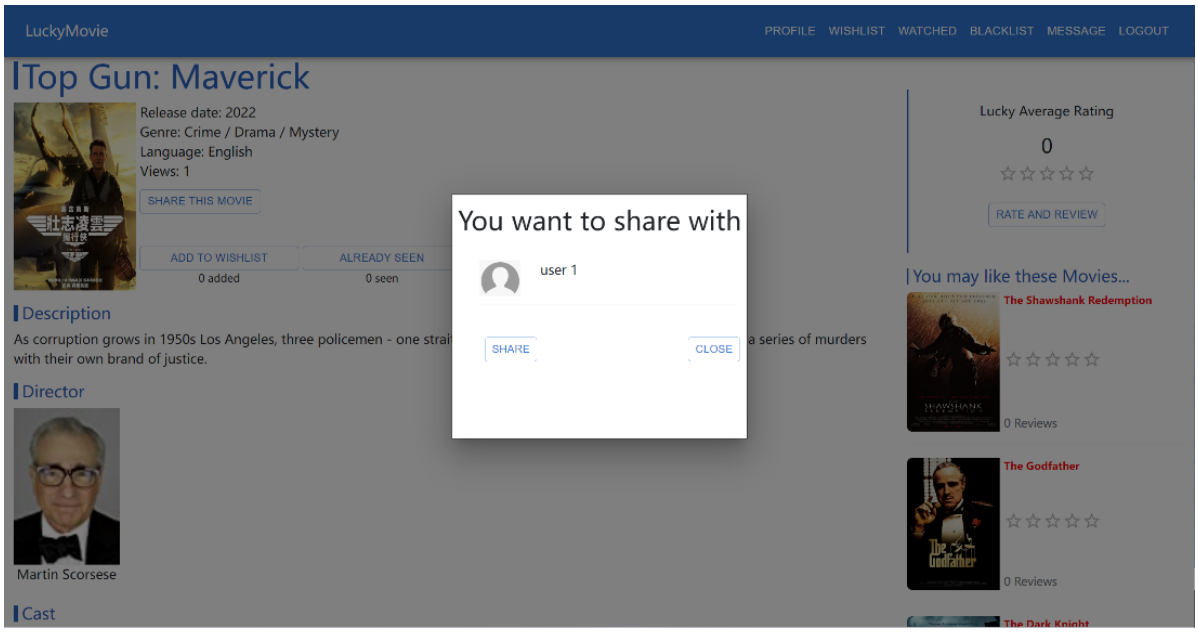
Condition	**Result**
Click any button	When you are not logged in, clicking any button will pop up a window to let you login
Click "Ratings and review"	It will show you the stars to rate and a box to enter your review
Click Actor's Avatar	Jump to Actor's page
Click Recommended movie posters	Jump to other movie pages
Click Director's Avatar	Jump to Director's page
Click "SHARE THIS MOVIE"	A window will pop up, allowing you to choose a friend to share this movie
Click "ADD TO WISHLIST"	Add this movie to your wishlist, then the button become "REMOVE FROM WISHLIST"
Click "ALREADY SEEN"	Add this movie to your watchedlist, then the button become "NOT ALREADY SEEN"



Movie review Example



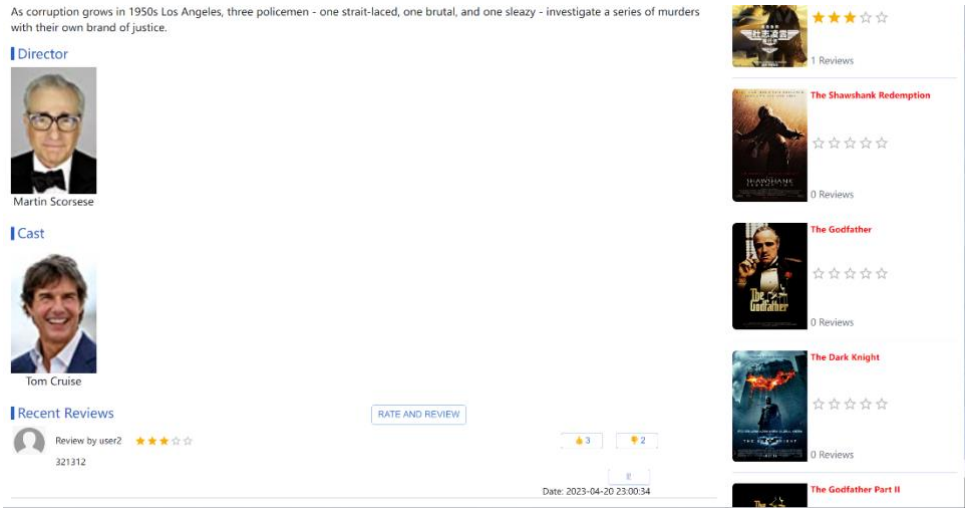
Share movie Example



2.4.1 Movies recent review

For the reviews on the movie page, users can like, dislike and delete their own reviews.

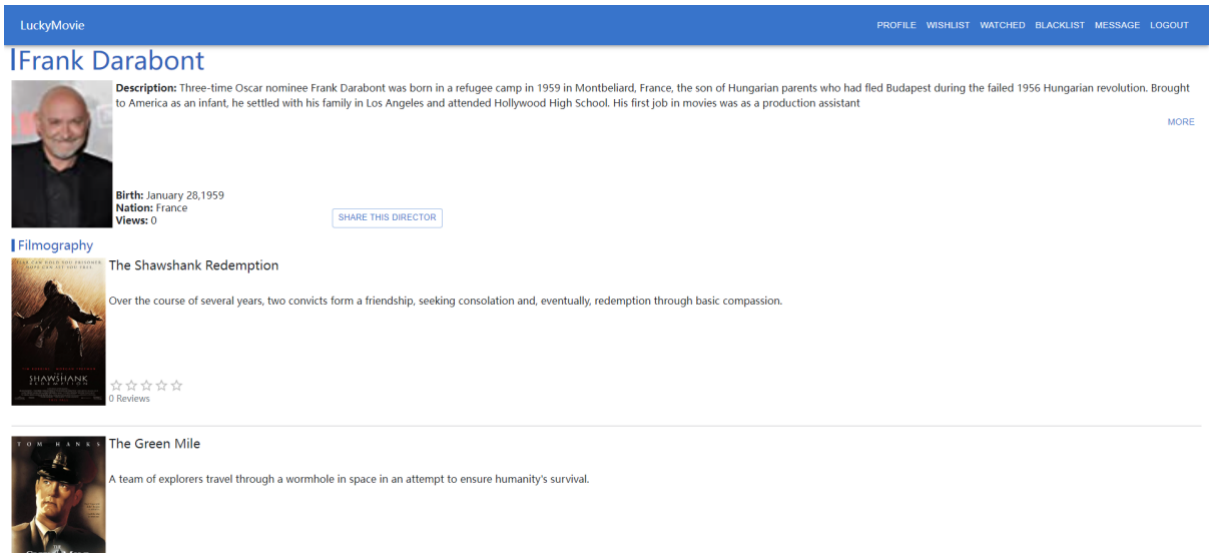
Conditions	Results
Click the like button	The like number will add one
Click the dislike button	The dislike number will add one
Click the trash can button	It will delete the current review



2.5 Director Page

Display the specific information of the director, age, nationality, appearance, and a brief description of the director.

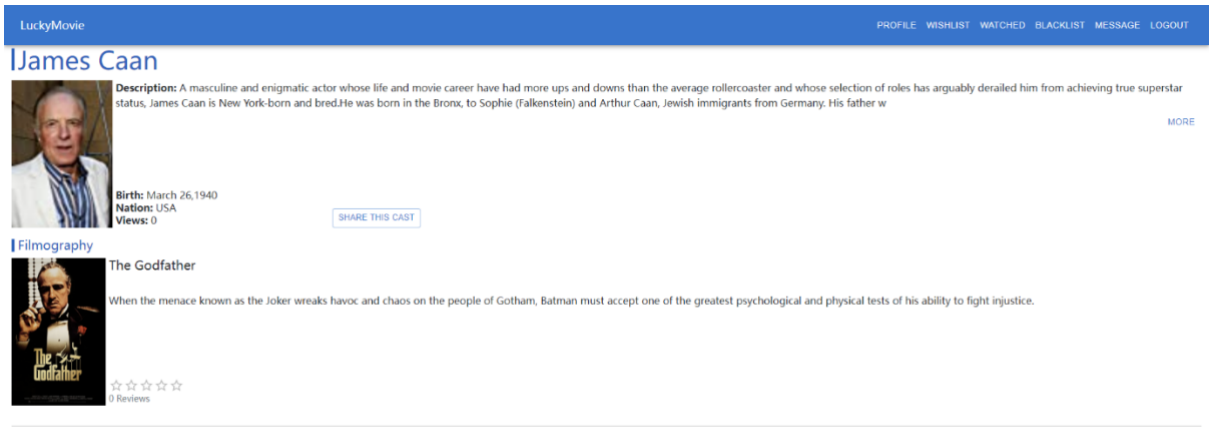
Conditions	Results
Click any button	When you are not logged in, clicking any button will pop up a window to let you login
Click ”SHARE THIS DIRECTOR” button	A window will pop up, allowing you to choose a friend to share this director
Click the movie posters	Jump to the movie detail page
Click “more”	Will show the whole description



2.6 Cast Page

Display the specific information of the cast, age, nationality, appearance, and a brief description of the cast

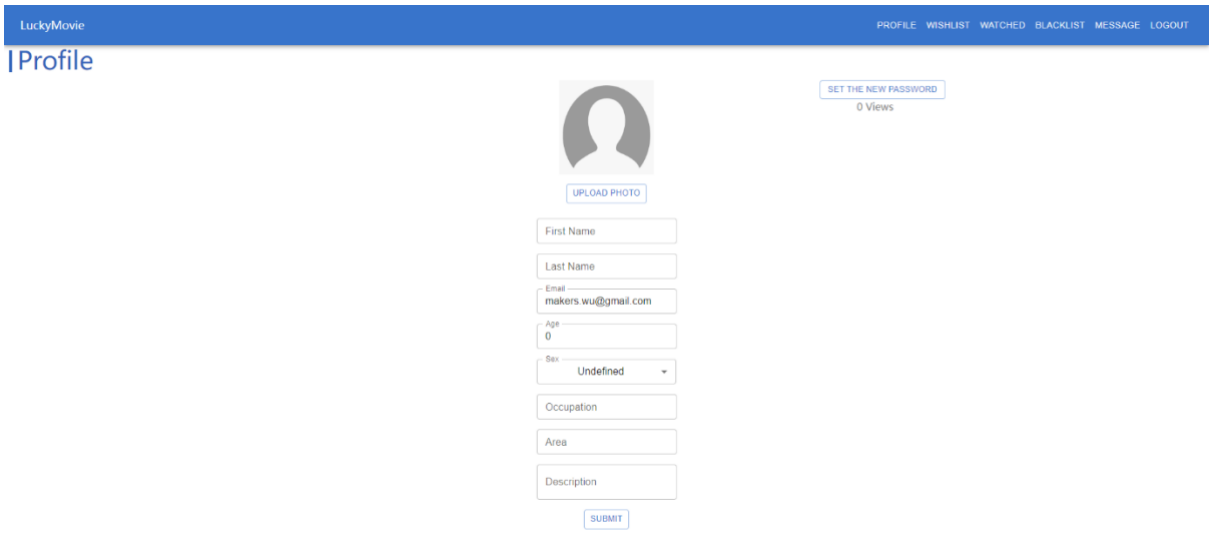
Conditions	Results
Click any button	When you are not logged in, clicking any button will pop up a window to let you login
Click ”SHARE THIS CAST” button	A window will pop up, allowing you to choose a friend to share this cast
Click the movie posters	Jump to the movie detail page
Click “more”	Will show the whole description



2.7 User Profile

Displays information about the current user.
You can modify your information on this page at any time, including uploading pictures and modifying first name, last name, age, sex, occupation, area and description.
After modifying, click the submit button, and your modified content will be automatically saved.

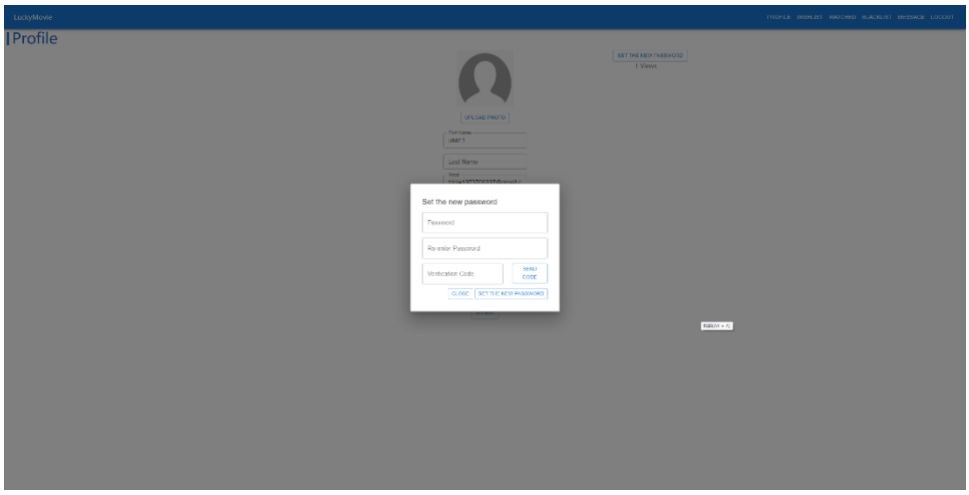
Condition	Result
Click “SUBMIT”	Successfully stored the information you entered into the database



2.7.1 set new password in profile page

You can also set a new password in the profile, and you need to accept email verification to set a new password.

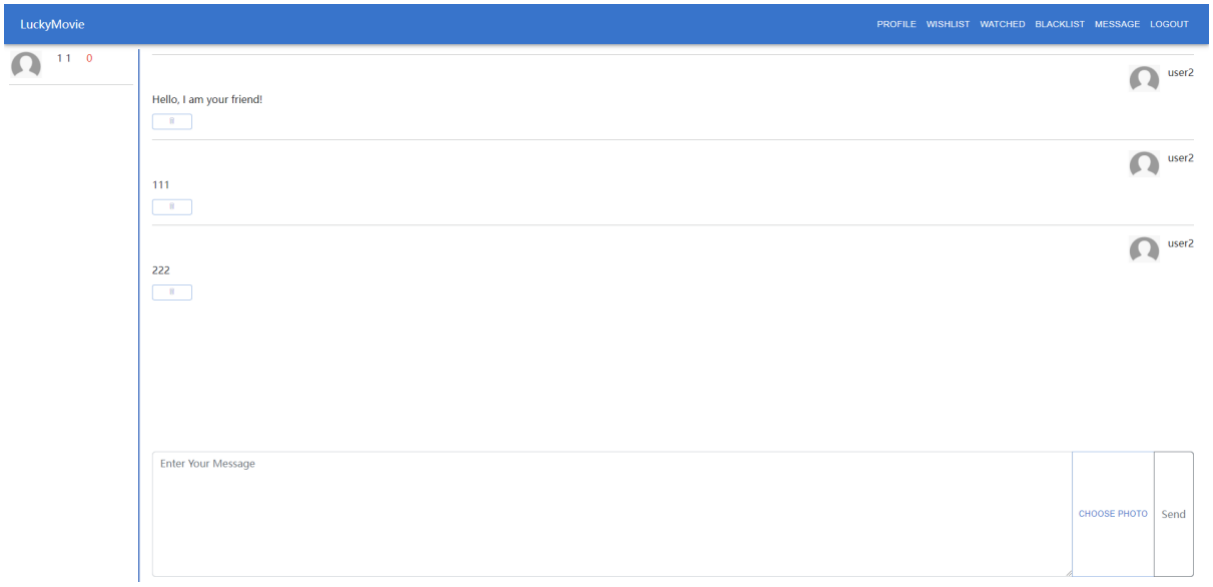
Conditions	Results
Click the “Send Code”	It will send a message including the vertification code to your email box
Click the “Set the new password”	This will change your password, An error will be reported if the passwords entered twice are inconsistent



2.8 Message Page

Our message page can get new messages in real time. The number of unread messages is displayed to the right of the user's name. You can test this function by opening another incognito browser and registering a new account.

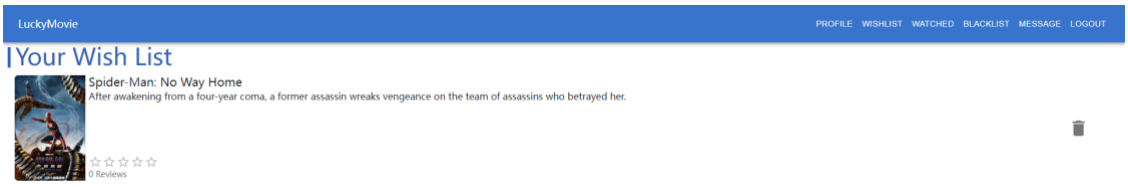
Conditions	Results
Click “Send” button	After modifying, click the submit button, and your modified content will be automatically saved.
Click ”CHOOSE PHOTO ”	You can choose the picture content to send.
Click Trash button below each message	Delete current message



2.9 Wish List Page

This page shows the list of movies you want to watch. Click the trash can button, you can delete the current movie.

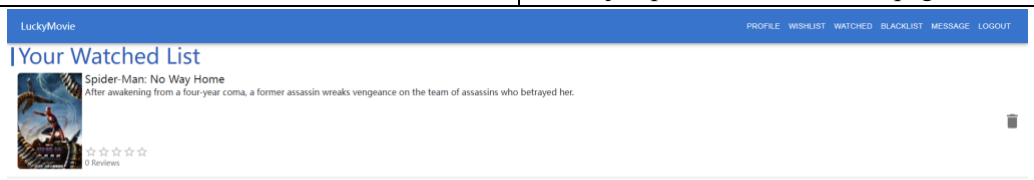
Condition	Result
Click the movie	It will jump to this movie detail page



2.10 Watched List Page

This page shows the list of movies you already watched. Click the trash can button, you can delete the current movie.

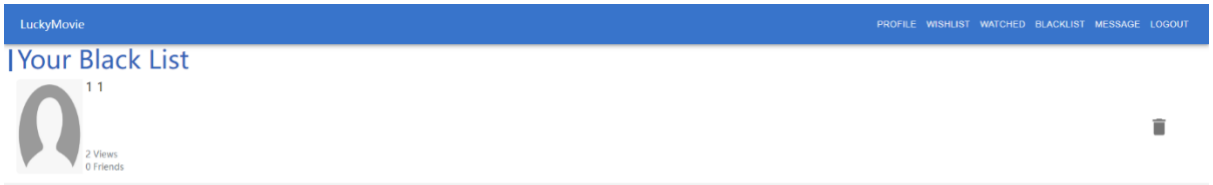
Condition	Result
Click the movie	It will jump to this movie detail page



2.11 Blacklist Page

This page shows the list of users you have blocked. You will not see reviews from or receive messages from these users. Click the trash can button, you can delete the user from the list. After that, you can see the user's reviews and communicate with him normally.

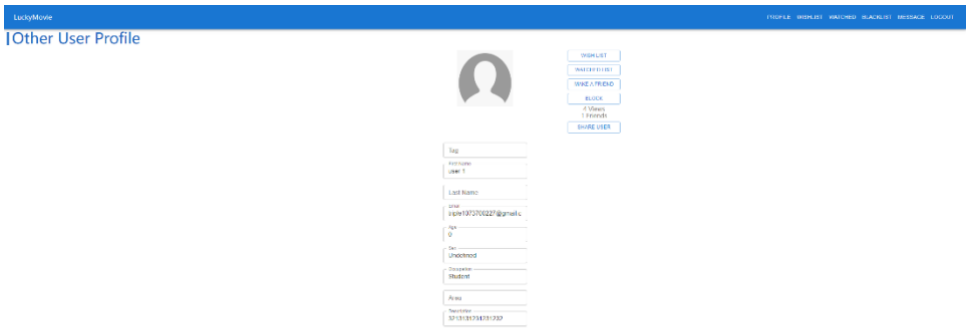
Condition	Result
Click the user	It will jump to this user’s profile page



2.12 Other profile

User can see other user’s tag, first name, last name, email, age, sex, occupation, area and description from this page.

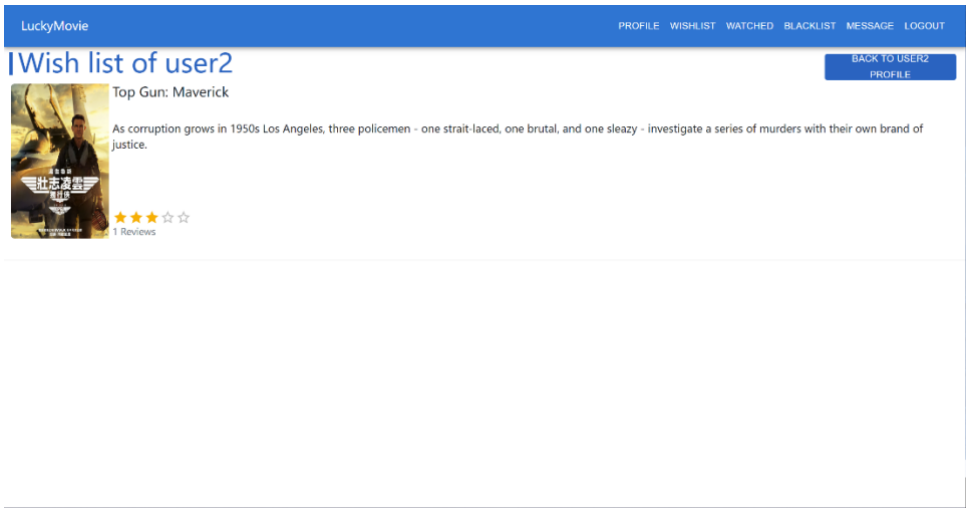
Conditions	Results
Click the “wish list”	It will jump to this user’s wish list page
Click the “watched list”	It will jump to this user’s watched list page
Click the “make friend”	You will make a friend with this user, and then you can chat with him in the message page
Click “block”	It will add this user in the blacklist, then you will not see this user’s reviews. Also you can delete this user from the blacklist
Click “share user”	A window will pop up, allowing you to choose a friend to share this user



2.12.1 Other user wish list

User can see the watchlist of other user’s wish list from this page

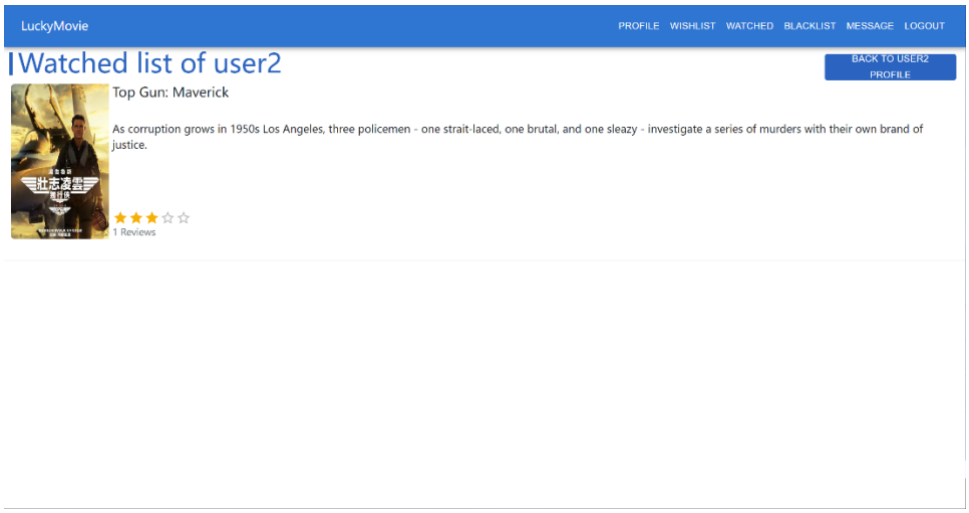
Conditions	Results
Click the “back to profile ”	It will return to this user’s profile page
Click the movie poster	It will jump to the movie detail page



2.12.2 Other user watched list

User can see the watchlist of other user's watched list from this page

Conditions	Results
Click the "back to profile "	It will return to this user's profile page
Click the movie poster	It will jump to the movie detail page



3.Technology Descriptions

3.1 Frontend

Our frontend is built using the ReactJS framework - a free and open-source library for creating user interfaces through UI components. ReactJS is one framework that enables our website as a
Single Page Application, which loads the majority of resources such as HTML, CSS and JavaScript at the beginning of our application's lifespan, which provides a quick and responsive user experience after the initial page load.
We have also chosen to utilize TypeScript, a superset of typed JavaScript to better manage our project while enabling static verification through strong static typing. This results in more robust software and allows for quicker development resulting from less time spent on debugging type issues.

3.1.1 MUI

MUI (Material-UI) is a popular React UI library that provides pre-built components and styles based on Google's Material Design guidelines. It is designed to help developers quickly and easily create responsive and visually appealing user interfaces for their React applications.
MUI components are highly customizable and can be easily styled to fit the look and feel of your application. The library includes a wide range of components, such as buttons, forms, modals, navigation menus, and more.
In the entire frontend development, we have used it extensively to create various components such as navigation design, button design, and more. This makes it very convenient for us to implement page navigation functionality.

3.1.2 react-bootstrap

React-Bootstrap is a popular UI library for React that provides pre-built components based on Bootstrap 4, a popular CSS framework. It allows developers to quickly and easily create responsive and mobile-first user interfaces for their React applications.
React-Bootstrap provides a wide range of components such as navigation bars, modals, forms, buttons, and more, which can be easily customized to fit the look and feel of your application.
We used this package to create navigation bars and search boxes, which simplified the handling of our web pages.

3.1.3 react-router-dom

React Router DOM is a package that provides routing capabilities to React applications. It enables developers to create dynamic, client-side routes in their applications, allowing users to navigate between different pages without the need for a full page reload.
The package is well-documented and regularly updated, with a large and active community of developers contributing to its development. Overall, React Router DOM is a great choice for developers who want to create powerful and flexible routing functionality in their React applications.
With the help of this package, we successfully designed multiple routes that prevented errors in page navigation. By combining this functionality with the previously mentioned button features, it became very convenient to navigate between web pages.

3.2 Backend

3.2.1 Flask

Flask is a lightweight Python web framework based on the Werkzeug toolkit and Jinja2 template engine. Its design philosophy emphasizes simplicity, flexibility, and ease of extension. Unlike some large frameworks, Flask is easy to learn and use, without excessive abstraction layers or dependencies. The framework was chosen as the back-end members were familiar with using Python. The Flask framework is widely used in various scenarios and can be used to develop web applications of any size, from small personal projects to large-scale enterprise applications.

3.2.2 Python libraries

3.2.2.1 Scikit-learn

Scikit-learn is a popular open-source machine learning library for Python. It provides a range of supervised and unsupervised learning algorithms, as well as tools for data preprocessing, model selection, and evaluation. The library is built on top of NumPy, SciPy, and matplotlib, and is designed to be easy to use and accessible to both novices and experts in machine learning.

We use this to carry out movie similarity calculations. Its cosine_similarity and countbectorizer helps as to calculate very easily. The recommendation system mainly based on it.

3.2.2.2 Pymysql

PyMySQL is a Python library used for working with MySQL databases. It is a lightweight and easy-to-use library that provides a simple interface for connecting to and querying MySQL databases from Python code. PyMySQL is a powerful and flexible library that is widely used for working with MySQL databases in Python applications.

We use it for the backend database connecting, querying and managing. It is easy to deal with the database selecting and executing.

3.2.2.3 JWT

The Python library for JSON Web Tokens (JWT) is a third-party library that provides functionality for creating, encoding, and decoding JWTs in Python. The library supports a variety of algorithms for encoding and decoding JWTs, including HMAC, RSA, and ECDSA.

It is very good to provide us with a method for user login encryption. We use jwt encode to encrypt the combination of user ID, session ID, and secret.

3.2.2.4 Pandas

Pandas is a popular open-source data analysis library for Python. It provides a powerful and flexible toolset for manipulating and analyzing tabular data, including data input/output, data cleaning, data transformation, data visualization, and statistical analysis.

We use pandas to read the crawled xlsx file and create a dataframe object. Various operations in pandas are used to process dirty data, such as drop, dropna, explode, etc.

We also use pandas to generate some xlsx files that we need. For example, in the movie recommendation section, we need a certain format of dataframe, so we use pandas to generate the specified format of xlsx file and then read it.

3.2.2.5 Pytest

Pytest is a popular Python testing framework that provides a simple and easy-to-use platform for writing and executing automated tests. It is designed to be scalable and extensible, making it suitable for a wide range of testing tasks, from unit tests to functional and integration tests.

Pytest offers a number of advantages over other testing frameworks, including its easy-to-learn syntax, flexible fixture system, and comprehensive reporting capabilities. Its use of standard Python asserts and the ability to integrate with other Python testing tools and libraries make it a popular choice for developers and testers alike.

When we write backend code, we always write tests first before writing the actual running code. This approach can both determine the functionality that the running equation needs to achieve and debug more quickly. We all know that without a test program, debugging the backend requires testing after completion of frontend work. This greatly reduces our coding efficiency. Thus, write pytest is necessary before actual coding in our project.

3.2.2.7 Hashlib

Hashlib is a Python library that provides a collection of cryptographic hash functions, which are used to convert arbitrary data into a fixed-size output, also known as a hash value or message digest. These hash functions are widely used in security and authentication applications, such as password verification, digital signatures, and data integrity checks.

We use this package together with jwt to encrypt user login. This package belongs to the third-party package with a high level of encryption, so that user privacy can be better protected.

3.2.2.8 datetime

Datetime is a Python library that provides a flexible and robust way to work with dates, times, and time intervals. It is a standard library in Python, and comes pre-installed with the language. The date class represents a date, with attributes for the year, month, and day. The time class represents a time of day, with attributes for the hour, minute, second, and microsecond. The datetime class represents a date and time together, with attributes for both the date and time components. The timedelta class represents a duration or interval between two dates or times.

The time provided by this package is very accurate and very easy to reference. So, we use this package to allow users to know the exact time when chatting with friends.

3.2.2.9 Json

Python's json library is a built-in module that provides functionality for encoding and decoding JSON data. JSON, which stands for JavaScript Object Notation, is a lightweight data interchange format that is widely used in web applications and APIs.

The json library provides two main functions: json.dumps() and json.loads(). The dumps() function is used to encode Python data structures into JSON format, while the loads() function is used to decode JSON data into Python data structures.

As a common package in the network framework, it is very convenient to transfer front-end and back-end data.

3.2.2.10 Sys and OS

The sys and OS libraries are two standard libraries in Python that provide access to system-specific functionality. These libraries are widely used in a variety of applications to interact with the operating system and perform system-level tasks.These two libraries make it easier for us to call local files.

3.2.2.11 flask_caching

Flask-Caching is a third-party library that provides caching support for Flask web applications. It allows developers to easily add caching functionality to their Flask applications by providing a simple and flexible caching interface.

Flask-Caching supports several caching backends, including in-memory caching, file-based caching, and Redis caching. It also provides several caching strategies, such as time-based caching, memoization, and key-based invalidation.

This package allowed us to successfully cache some repeated requests and speed up the webpage. Enable users to get a better experience.

4. Implementation challenges

4.1 Frontend

4.1.1 ReactJS

First, we attempted to code purely in JavaScript, but we found that this led to a lot of front-end coding tasks. Therefore, we decided to learn ReactJS to improve front-end coding efficiency. However, this was a challenge for some team members who were not proficient in ReactJS. ReactJS allows for the creation of reusable UI components, which improves code modularity and maintainability. In actual coding, we also generated many reusable UI components, making our code more modular and easier to maintain.

4.1.2 UI design

Our initial UI design was not aesthetically pleasing, so we searched online for Material-UI and React-Bootstrap, two UI frameworks based on ReactJS. We used many ready-made components, such as App Bar, Rating, Modal, Button, Textfield, and so on. These ready-made component UI designs are very beautiful and easy to understand. Now our website design is very beautiful.

4.1.3 frontend sends requests to the backend

At the beginning of coding, we directly used the fetch function to get the response and then assigned it to use it. But we found that no matter how we assigned it, the variable still seemed to not have been assigned. After researching fetch in ReactJS online, we found that fetch is an asynchronous method, which means that assigning before getting the response is invalid, even if the assignment operation is written after fetch. So we wrapped fetch into a promise and wrote it as a helper function. Promises can ensure that the assignment operation is performed after getting the response.

4.1.4 Save scroll bar position

Initially, the scroll bar on our message page always automatically slid to the bottom. We want to ensure that the scroll bar on the Message page does not change during polling of the server, but automatically scrolls to the bottom when there is a new message. We used the useRef to save the position of the scroll bar. Each time the React component is re-rendered, the reference created by useRef is not re-assigned. We also compare the retrieved Message List with the current Message List after each poll of the server by using JSON.stringify. If the retrieved Message List is the same as the current Message List, we do not perform any operations. This ensures that there is no re-rendering and speeds up response time.

4.1.5 Login status

Different browser windows share the user's login status. Initially, when we completed Login and Sign Up, we found that the user's login status could not be shared in different browser windows. After searching online, we found that localStorage can store the user's login status in the browser. localStorage ensures that we can share the user's login status in different browser windows.

4.1.6 Upload Picture

At the beginning, we were unsure of how to save images uploaded by users to our server. After conducting research, we discovered that we could encode the uploaded images as a Base64 string and send them to the server for storage. To assist with encoding the images as Base64, we wrote a helper function called "filetoDataUrl". This method reduced the size and time required for image data transmission and ensured secure transmission.

4.2 Backend

4.2.1 Database Selection

Initially, our choice was MongoDB, which is a database based on distributed file storage. It is

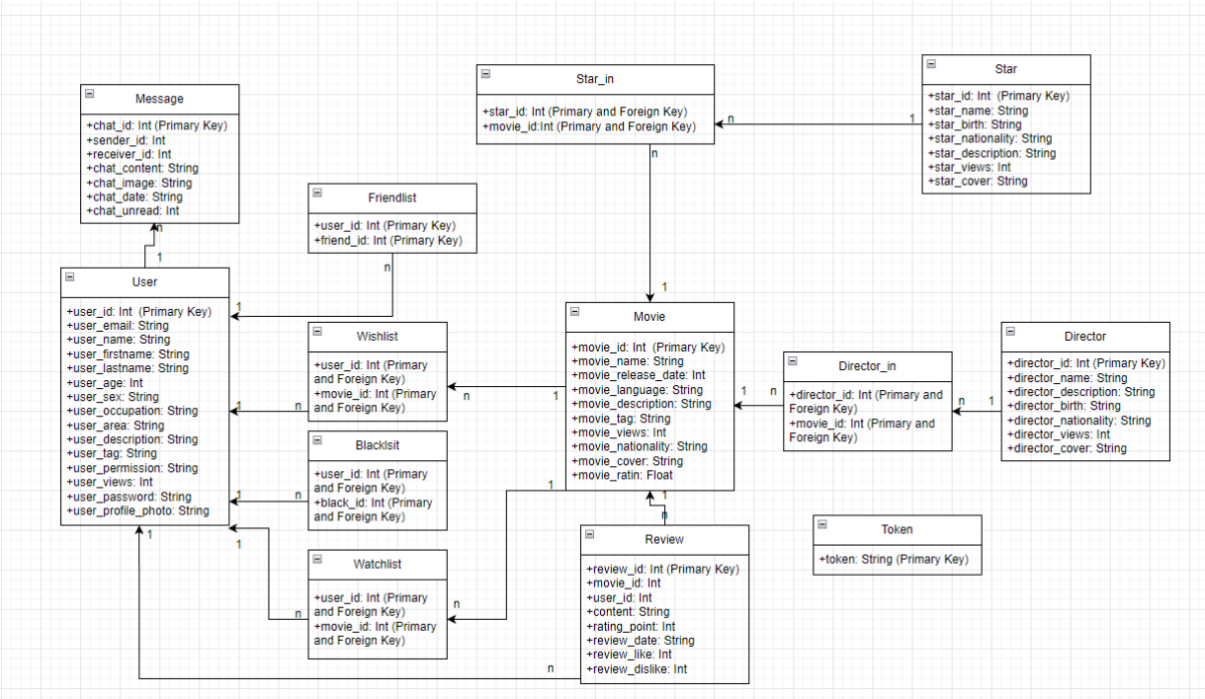
a product between a relational and non-relational database. Since it is not a familiar relational database for us, it was difficult to learn at the beginning. As we could not confidently handle dirty data and establish relationships between various data during exploration, we turned to using MySQL.

4.2.2 Data Acquisition

When we learned that the IMDb API required payment and approval, we gave up using open-source APIs and instead used web crawlers to obtain basic information for some movies. During the crawling process, the data was often contaminated, like the string is often truncated, obfuscated, or deleted. We used pandas to process them one by one, making the data readable on our website.

4.2.3 Database Design

Database design issues can only be discovered in actual work. To make accessing data more efficient, we have made several adjustments to the database after submitting proposals. The main changes are reflected in adjustments to data types and adjustments to many-to-many relationships. In the actual coding process, we found that some data types were not long enough, so we had to make the data types can contain more data length. The adjustment of many-to-many relationships was made after data crawling process. Since a movie does not necessarily have only one director and one lead actor, we had to expand the original one-to-many relationships into one-to-many and many-to-one relationships to achieve the effect of many-to-many.



4.2.4 Database Connection

During the demo B, we discovered efficiency issues when connecting the database to the backend code. During the demonstration of the message function, we found that a large number of rapid requests would cause our system to crash. Since our message reminder function is

implemented by continuously sending requests at intervals, we needed to solve the problem of system crashes. After researching, we chose to use Pymysqlpool to establish multiple connections before the server start to avoid too many connections per request.

4.2.5 Framework Selection

We initially chose the lightweight web application framework Flask. Later, we wanted to switch to the advanced Flask-RESTful but gave up. Several of our team members took the course COMP9321 Data Services Engineering this semester, where Flask-RESTful was mentioned until the midterm, so we only realized we should use a more advanced flask at that time. However, changing the framework at that point required too much work, so we gave up. But we tried to standardize each API and function as much as possible, like writing clear comments before each function.

4.2.6 Website Security

Encryption is performed whenever sensitive data needs to be transmitted. We use generate_new_session_id and HS256 to encrypt the identification of each logged-in user. We also use sha256 to encrypt user passwords. These two measures are taken to prevent user information from being stolen. Email verification is our measure to prevent massive malicious registration. Each email can only register one account, and each account needs to be verified by email, which is the most effective two methods in my opinion. Setting a cache mechanism for specific pages can also prevent system crashes caused by a large number of accesses, which is also related to security.

4.2.7 Recommendation System

We believe that a recommendation system should consist of two parts. One part consists of movies that are similar, and the other part consists of movies that other users who liked this movie also liked.

For similar movies, we tried many approaches. We tried using the TF-IDF vectorizer to vectorize movie descriptions, using scikit-learn and NLTK libraries to annotate features and calculate similarities, using the Surprise library to read datasets and train recommendation models with the SVD algorithm, and implementing a deep autoencoder with TensorFlow and Keras. Finally, we chose to use a cosine similarity matrix that combines movie tags, director names, and actor names, and then compare them through pairwise similarity scores between the given movie and all other movies. Our system also performs additional calculations when the user is logged in. This includes removing some recommended movies that the user has already watched, added to their wish list, or commented on.

For the part where we recommend movies that the user might like, we add up all the wish lists of users who have added this movie to their wish list, and then we select the highest rated movie from these movies to show to the user. We believe that by leveraging users with similar tastes, we can find movies that the user may potentially like.

4.2.8 Unique Search System

We believe that the purpose of a search system is to allow users to quickly find what they are looking for. So, we think that mandatory inclusion and mandatory exclusion can improve search efficiency. We achieve this by searching for content between "", -"", "" and -"".

5.User Documentation

5.1 As an explorer

As an explorer in a movie recommendation system, you have the opportunity to broaden your movie-watching horizons. The system can suggest films that you might like. This means that you can discover hidden gems or explore genres and subgenres that you haven't yet explored.

Moreover, the recommendation system can provide a personalized movie-watching experience by tailoring its suggestions to your unique tastes. It can take into account factors like your wish list, watched list, and reviewed movie. This can save you time and effort in searching for new movies to watch, as the system can do the work for you.

In addition, the movie recommendation system can provide recommendations from other users with similar interests. This can help you discover movies that you might not have found on your own and can lead to interesting discussions and recommendations.

Overall, the movie recommendation system can provide a rich and diverse movie-watching experience, allowing you to explore new genres, discover new favorite movies, and connect with other movie lovers.

5.2 As a contributor

As a contributor to a movie recommendation system, you can contribute various things such as:

Rating and reviewing movies: By rating and reviewing movies you've watched, you can help the recommendation system understand your preferences and suggest movies that you may like.

Providing detailed feedback: If you have specific feedback about a movie or a particular aspect of a movie, you can share that feedback with the recommendation system. This can help the system improve its recommendations and provide better suggestions to other users.

Share unique perspectives: You can share your unique perspectives with the friends made on the websit.

Overall, by actively participating and providing feedback in a movie recommendation system,

you can help improve the system's accuracy and provide better movie recommendations for all users.

6.User Manual (Lubuntu 20.4.1 LTS virtual machine image in the VirtualBox 6.1.44)

Remember!!! Everytime open a new terminal you always run the code:

```
$ sudo su
$ cd
```

6.1Checking the MySQL

Firstly, check if the computer has the MySQL server. Open a terminal. Use the code:

```
$ mysql -V
```

If no version is displayed, it means that there is no MySQL on this computer.

Then you need to install the MySQL.

1. Update package list and upgrade packages:

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

2. Install the MySQL package:

```
$ sudo apt-get install mysql-server
```

3. During the installation, you will be prompted to set a root password for MySQL. You should set the password as ‘123456’.

```
$ ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY '123456';
```

4. When installation was done, you can use the MySQL by using the command:

```
$ mysql -uroot -p
```

Then enter ‘123456’

5. Create the database by running:

```
$ create database MOVIEFINDER;
```

6.2 Checking the Nodejs

Firstly, we need to install nodejs by entering:

```
$ sudo apt install nodejs
$ sudo apt install npm
$ sudo npm i -g yarn
```

After these progress, you should check the nodejs version by entering:

```
$ node -v
```

If the version not >= 16.14.2

You should run:

```
$ npm install -g n
$ n stable
```

Then close the terminal, open the terminal again. Now check the nodejs version again.
It will change to newest version.

6.3 Checking the Python libraries.

Before starting the backend, you should check if the python libraries is all installed.
Just run:

```
$ cd ~/capstone-project-3900h18bluckyteam
$ pip install -r requirements.txt
```

Then it will be ok.

6.4 Starting the Backend:

Open a new terminal, clone the project repository to the directory.
~/capstone-project-3900h18bluckyteam.
Next, use the commands:

```
$ cd ~/capstone-project-3900h18bluckyteam/backend/src
$ python3 server.py
```

It will start the backend.

6.5 Starting the Frontend:

Once the backend is running, open a new terminal.
run the frontend server by entering the next set of commands:

```
$ cd ~/capstone-project-3900-h18b-luckyteam/frontend
$ yarn install
$ yarn start
```

If you firstly using the code, you should enter the

```
$ yarn install.
```

Then every time you use it, you just need to type:

```
$ yarn start.
```

At last, open the Firefox browser, go to the “http://localhost:3000”, then you can use the project well.

6.6 Emergency

When you newly open the backend and frontend, you will find that the word logout appears in the upper right corner of the page, and you cannot log out. In this case, the logout operation was not performed after the backend was forced, resulting in token invalidation. The error shows in Figure 6.5.1

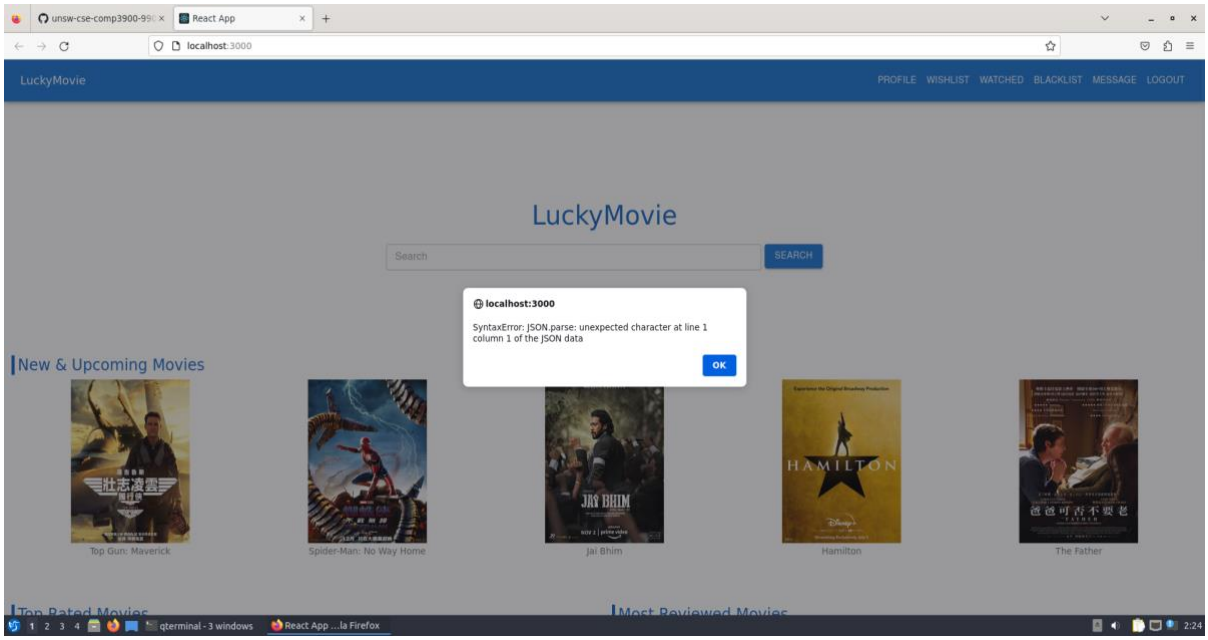


Figure 6.5.1

At this time, you need to open the browser console as press F12 on the keyboard.
Then typing “localStorage.removeItem('luckyToken')” in the console and click the enter button in the keyboard. As Figure 6.5.2

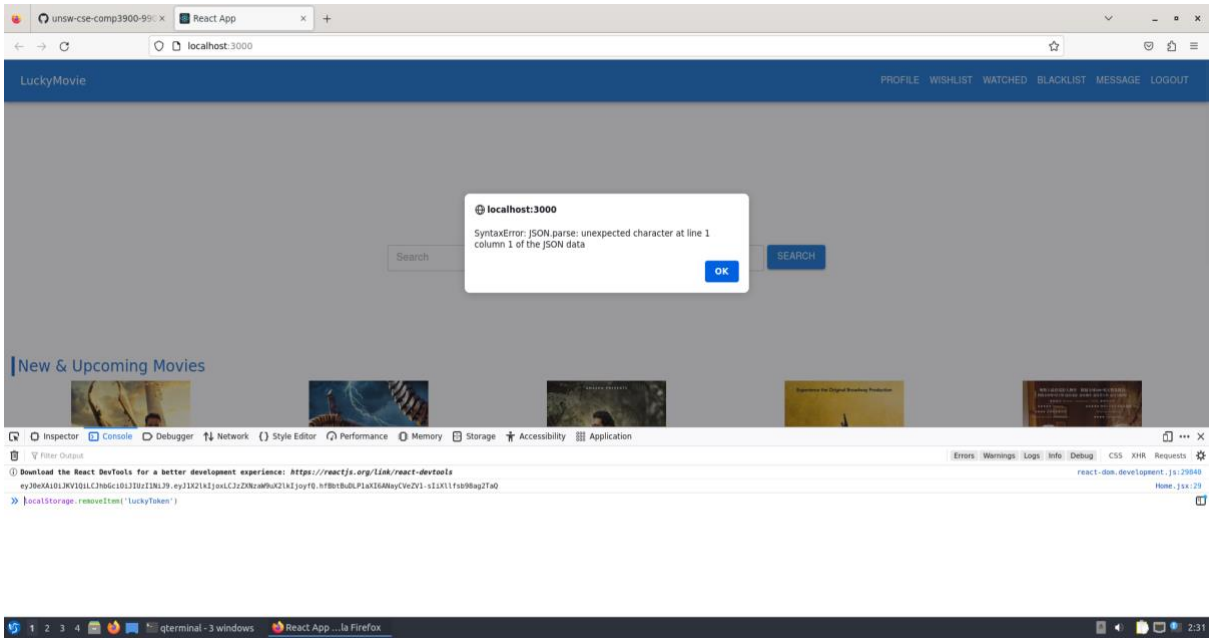


Figure 6.5.2

Then refresh the page and it will work normally.

7 Reference

Moon, S. (2020). Effects of COVID-19 on the Entertainment Industry. *IDOSR Journal of Experimental Sciences*, *5*(1), 8-12.

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